Connor L. Brown

Connor.Brown@KU.edu | 409-434-8520
0000-0002-9680-8930 | ConnorB.github.io

Research Interests

Stream ecosystem ecology and biogeochemistry; carbon cycling, ecosystem metabolism, and oxygen dynamics in non-perennial streams; hydrologic, land use, and global change effects on aquatic ecosystem function.

Education

Ph.D., Geology 2021–Present
University of Kansas, Lawrence, Kansas
Relevant Coursework: Environmental Data Analysis and Statistics, Environmental Geophysics, Geomorphology, Stream Tracers, Water Quality Modeling, Team Science, Biogeochemistry

M.Sc., Biology 2019–2022
Sam Houston State University, Huntsville, Texas
Thesis: Ecosystem Metabolism of Coastal Texas Streams Across Precipitation Regimes and Land Use Gradients
Relevant Coursework: Ichthyology, Invertebrate Zoology, Biogeography, Biostatistics, Hydrology, Stream Ecology

B.Sc., Natural Resources Management 2016–2018
Texas Tech University, Lubbock, Texas
Concentration: Aquatic and Fisheries Biology
Minor: Geographic Information Science and Technology
Relevant Coursework: Freshwater Bioassessment, Watershed Planning, Fisheries Conservation and Management, Quantitative Methods in Natural Resources, Spatial Analysis

Research Experience

Graduate Research Assistant 2021–Present
Kansas Geological Survey, University of Kansas
Conducting dissertation research on the effects of streamflow intermittency on ecosystem metabolism, dissolved oxygen dynamics, and carbon cycling in non-perennial streams. Contributed to the NSF EPSCoR-funded AIMS (Aquatic Intermittency effects on Microbiomes in Streams) project, a highly interdisciplinary team of hydrologists, biogeochemists, microbiologists, and community ecologists characterizing the effects of intermittency on biogeochemical cycling and macroinvertebrate and microbial community composition and function across sites in the Great Plains, Mountain West, and Southeast Forest regions. Collaboratively designed and helped implement a reach-scale experimental stream drying and

rewetting study at Konza Prairie Biological Station, including water quality sampling, dissolved gas sampling, macroinvertebrate community sampling, and designing and deploying high-frequency water quality sensors.

Graduate Research Assistant 2019–2021
Department of Biological Sciences, Sam Houston State University
Managed data collection, analyzed time series data, estimated ecosystem metabolism using R, and collected macroinvertebrates and fish for the NSF-funded project Thresholds in Ecosystem Responses to Rainfall Gradients (TERRG). Estimated daily gross primary production, ecosystem respiration, and gas exchange, and applied structural equation modeling to relate precipitation and land use gradients to stream metabolism. Collected and processed water samples for dissolved organic carbon and nutrient analysis.

Graduate Research Assistant 2019
Texas Research Institute for Environmental Studies, Sam Houston State University
Worked on a funded grant from the Texas Military Department to collect invertebrate and habitat data, identified terrestrial and aquatic invertebrates, and pinned specimens.

Undergraduate Researcher 2016–2018
Department of Natural Resources Management, Texas Tech University
Proposed and conducted a research project on the Pecos River. Field work included deploying Hester-Dendy samplers and collecting habitat measurements. Lab work included sorting and identifying macroinvertebrates using dichotomous keys.

Undergraduate Internship 2016
Department of Natural Resources Management, Texas Tech University
Assisted with dissertation-related field and lab work, including transect setup, macroinvertebrate and fish collection, water sampling, habitat measurements, macroinvertebrate identification, and environmental DNA extraction.

Teaching Experience

Teaching Assistant, Field Ecology Fall 2024, Fall 2025
Environmental Studies Program, University of Kansas
Assisted with a field-based course introducing research methods in environmental science, including data collection and analysis across lake, stream, wetland, and prairie ecosystems. Supported lab sections covering ecosystem sampling, aquatic invertebrate identification, GIS, and introductory R, and helped students with data analysis and interpretation assignments.

Teaching Assistant, Senior Capstone Spring 2025, Spring 2026
Environmental Studies Program, University of Kansas
Supported semester-long, interdisciplinary capstone research projects in which student teams partnered with campus and community stakeholders on real-world environmental issues. Assisted with project management and professional communication workshops, met with student groups to advise on research progress, and supported final written reports and oral presentations.

Teaching Assistant, General Ecology — Spring 2019, Fall 2019

Department of Biological Sciences, Sam Houston State University

Led laboratory and field sections for an upper-division course covering organismal, population, community, and ecosystem ecology. Guided students through field-based application of ecological principles, lab exercises, and data collection and interpretation.

Teaching Assistant, Environmental Science — Fall 2019

Department of Biological Sciences, Sam Houston State University

Led lab sections for an introductory, non-majors course on contemporary environmental issues, including air, water, and soil pollution, biodiversity, climate change, agriculture, population growth, and energy. Assisted students with lab activities and core environmental science concepts.

Publications

1. Flynn, S. M., Hale, R., Seybold, E., Plont, S., Busch, M., Sommerville, A., **Brown, C. L.**, & Burgin, A. J. (2026). Network connectivity and local hydrology drive DOM composition in a non-perennial stream. *Journal of Geophysical Research: Biogeosciences*. doi:hb7kr8
2. Frazier, C. F., **Brown, C. L.**, Hamersky, M., Coole, T., Campobasso, M., Thomas, C., & Harris, T. D. (2025). Spatiotemporal variation of ecosystem metabolism in a eutrophic, polymictic reservoir: Water column stability begets metabolism stability. *Inland Waters*. doi:hb7kr9
3. Plont, S., Peterson, D. M., Smith, C. R., Bond, C. T., Tchamba, A. L. K., Wolford, M. A., Zarek, K., Speir, S. L., Jones, C. N., Benstead, J. P., Busch, M. H., Hale, R. L., **Brown, C. L.**, Seybold, E. C., Shogren, A. J., Kuehn, K. A., You, Y., Jackson, C. R., Burgin, A. J., & Atkinson, C. L. (2025). Hydrologic, biogeochemical, microbial, and macroinvertebrate responses to network expansion, contraction, and disconnection across headwater stream networks with distinct physiography in Alabama, USA. *Earth System Science Data Discussions*. doi:hb7ksb
4. Groff, C. M., Kinard, S. K., Hogan, J. D., Whiles, M. R., Ulseth, A. J., Strickland, B. A., Carvallo, F., Jenkins, V., Solis, A. T., Groff, D. A., Frazier, C., **Brown, C. L.**, & Patrick, C. J. (2025). Precipitation regime drives key alterations in subtropical stream fish community assembly and functional trait distribution across a major rainfall gradient. *Journal of Freshwater Ecology*. doi:hb7ksc
5. **Brown, C. L.**, Delaune, K. D., & Pease, A. A. (2024). Benthic macroinvertebrate assemblage structure in salinized reaches of the lower Pecos River, Texas. *The Southwestern Naturalist*. doi:hb7ksd
6. Delaune, K. D., Pease, A. A., Patiño, R., **Brown, C. L.**, & Barnes, M. A. (2024). Gulf killifish (*Fundulus grandis*) in the Pecos River: Unique life history traits in a nonnative, inland population. *The Southwestern Naturalist*. doi:rbkq

7. **Brown, C. L.**, Frazier, C. F., Patrick, C. J., & Ulseth, A. J. (in preparation). Ecosystem metabolism of coastal Texas streams across precipitation regimes and land use gradients. *Ecosystems*.

Published Datasets

1. Busch, M., Belskis, A., **Brown, C. L.**, Flynn, S., Cook, S., Allen, D., & Burgin, A. (2026). AIMS Great Plains, macroinvertebrate sequences (AIMS_GP_MACR). *HydroShare*. doi:hb7ksf
2. **Brown, C. L.**, Kraft, M., Hale, R., & Godsey, S. (2026). AIMS Johnston Draw continuous stage and discharge at watershed outlet data (AIMS_MW_JDR_DISC). *HydroShare*. doi:hb7ksg
3. **Brown, C. L.**, Kraft, M., Hale, R., & Godsey, S. (2026). AIMS Gibson Jack Creek continuous stage and discharge at watershed outlet data (AIMS_MW_GBJ_DISC). *HydroShare*. doi:hb7ksh
4. **Brown, C. L.**, Bilbrey, E., Lanfear, R., Godsey, S., & Hale, R. (2026). AIMS Johnston Draw continuous water quality at watershed outlet data (AIMS_MW_JDR_EXOS). *HydroShare*. doi:hb7ksj
5. **Brown, C. L.**, Lanfear, R., Bilbrey, E., Godsey, S., & Hale, R. (2026). AIMS Gibson Jack Creek continuous water quality at watershed outlet data (AIMS_MW_GBJ_EXOS). *HydroShare*. doi:hb7ksk
6. **Brown, C. L.**, & Seybold, E. (2026). Youngmeyer Ranch (KS) continuous discharge at watershed outlet data (AIMS_GP_YMR_approach1_DISC). *HydroShare*. doi:hb7ksn
7. **Brown, C. L.**, & Seybold, E. (2026). Shane Creek (Konza Prairie) continuous discharge at watershed outlet data (AIMS_GP_SHN_approach1_DISC). *HydroShare*. doi:hb7ksp
8. **Brown, C. L.**, & Seybold, E. (2026). Kings Creek (Konza Prairie) continuous discharge at watershed outlet data (AIMS_GP_KNZ_approach1_DISC). *HydroShare*. doi:hb7ksq
9. **Brown, C. L.**, Flynn, S., Burgin, A., & Seybold, E. (2026). AIMS continuous water quality at Shane Creek outlet (AIMS_SHN_approach1_EXOS). *HydroShare*.
10. **Brown, C. L.**, Zipper, S., & Busch, M. (2026). Youngmeyer Ranch, KS environmental data (AIMS_GP_YMR). *HydroShare*. doi:hb7ksr
11. **Brown, C. L.**, Utzman, C., Flynn, S., & Busch, M. (2026). Shane Creek (Konza Prairie) experiment stream temperature, intermittency, and conductivity data (AIMS_GP_SHN_approach4_STIC). *HydroShare*. doi:hb7ksm
12. **Brown, C. L.**, Flynn, S., Burgin, A., & Seybold, E. (2025). AIMS continuous water quality at King's Creek outlet (AIMS_KNZ_approach1_EXOS_V1.0). *HydroShare*.
13. **Brown, C. L.**, Flynn, S., Utzman, C., Busch, M., Sommerville, A., Wilhelm, J., Layman, T., Dorantes, C., Zeglin, L., Seybold, E., & Burgin, A. (2025). Shane Creek, field data (AIMS_GP_SHN_YSIS). *HydroShare*.

Conference Presentations

1. **Brown, C. L.**, Seybold, E. C., Plont, S., Hale, R. L., Benstead, J. P., & Burgin, A. J. (2026). Effects of drying frequency and duration on ecosystem metabolism in non-perennial streams. *Society for Freshwater Science*, Spokane, WA.
2. **Brown, C. L.**, Busch, M. H., Plont, S., Benstead, J. P., Burgin, A. J., Hale, R. L., & Seybold, E. C. (2025). Hydrologic controls on oxygen regimes in non-perennial streams. *Great Plains Limnology*, Manhattan, KS.
3. **Brown, C. L.**, Busch, M. H., Plont, S., Benstead, J. P., Burgin, A. J., Hale, R. L., & Seybold, E. C. (2025). Hydrologic controls on oxygen regimes in non-perennial streams. *Society for Freshwater Science*, San Juan, PR.
4. **Brown, C. L.**, Plont, S., Burgin, A. J., & Seybold, E. C. (2024). The oxygen regimes of non-perennial streams. *The Water Science Conference*, Saint Paul, MN.
5. **Brown, C. L.**, Burgin, A. J., & Seybold, E. C. (2023). Flow variability drives changes in the oxygen regime of a non-perennial prairie stream. *Kansas Governor's Water Conference*, Manhattan, KS.
6. **Brown, C. L.**, Wheeler, C. T., Speir, S. L., Allen, D. C., Benstead, J. P., Burgin, A. J., Hale, R. L., & Seybold, E. C. (2022). Duration of drying controls the magnitude and recovery of ecosystem metabolism in U.S. intermittent streams. *Kansas Governor's Water Conference*, Manhattan, KS.
7. **Brown, C. L.**, Wheeler, C. T., Speir, S. L., Allen, D. C., Benstead, J. P., Burgin, A. J., Hale, R. L., & Seybold, E. C. (2022). Characterizing metabolism dynamics in U.S. intermittent streams. *Joint Aquatic Sciences Meeting*, Grand Rapids, MI.
8. **Brown, C. L.**, Carvallo, F., Frazier, C., Groff, C. M., Jenkins, V., Kinard, S. K., Solis, A. T., Strickland, B. A., Hogan, J. D., Patrick, C. J., Whiles, M. R., Vander Zanden, H., & Ulseth, A. J. (2021). Ecosystem metabolism of coastal Texas streams across precipitation and land use gradients. *Society for Freshwater Science*, Virtual.
9. **Brown, C. L.**, Patrick, C. J., Hogan, J. D., Reese, B., Whiles, M. R., Vander Zanden, H., Carvallo, F., Frazier, C., Groff, C. M., Jenkins, V., Kinard, S. K., Solis, A. T., & Ulseth, A. J. (2020). Exploring the impact of land use on primary production biomass across Texas coastal streams. *ASLO–SFS 2020 Joint Summer Meeting (meeting cancelled)*.
10. **Brown, C. L.**, Delaune, K. D., & Pease, A. A. (2018). Spatial and temporal variation in benthic macroinvertebrate assemblage structure in salinized reaches of the Pecos River. *Desert Fishes Council*, Death Valley, CA.
11. **Brown, C. L.**, Delaune, K. D., & Pease, A. A. (2018). Macroinvertebrate community structure on the lower Pecos River. *Southwestern Association of Naturalists*, San Marcos, TX.

Student Mentoring

- Claudia Dorantes — Undergraduate, Lab Technician, University of Kansas (2024–2026)
- Taylor Layman — Undergraduate, Lab Technician, University of Kansas (2024–2026)
- Kari Snelding — Master's Student, University of Kansas (2022–2023)
- Dani Gray — Undergraduate, Sam Houston State University (2019–2021)
- Naomi Smith-Wade — Undergraduate, Sam Houston State University (2020–2021)
- Marlyn Hernandez — Undergraduate, Sam Houston State University (2021)

Service & Committees

- Kansas Geological Survey — Diversity, Equity, and Inclusion Committee (2023–2024)
- Society for Freshwater Science — Diversity, Equity, and Inclusion Committee (2021)
- Biological Sciences Graduate Student Organization, Treasurer (2019–2020)
- Texas Tech Gay-Straight Alliance (2016–2018)
- Circle K International, Texas A&M–Corpus Christi, President (2014–2016)

Professional Memberships

- Society for Freshwater Science (2018–Present)
- Desert Fishes Council (2018–Present)
- Southwestern Association of Naturalists (2017–Present)

Research Funding

- **Brown, C. L.** (2017). Center for Active Learning and Undergraduate Research Engagement, Texas Tech University ($1,000).

Certifications

- GeoApps Certification, Esri (2018)
- SSI Advanced SCUBA Certification (2009)